

Responsible AI Statement

Predicting Child Stunting Growth in Kenya

1. Who does this system affect?

Three groups. (a) Users who interact with the model's output: community health workers and clinicians at growth-monitoring sessions, who would see a severity flag alongside a child's record. (b) People the model makes decisions about: individual children under five sampled in the KDHS, most of whom will never know a model informed the dietary advice their caregiver received. (c) A group affected but without a voice in the design process: children in remote, pastoralist, and insecure areas, who are undercounted in DHS enumeration relative to their share of national stunting burden, and whose caregivers and community health structures were never consulted on how a model output should — or shouldn't — be used in their care.

2. What is the worst plausible outcome?

A child who is moderately stunted is classified "Normal growth" because the model leans heavily on household wealth and maternal education — strong population-level predictors that can still be wrong for an individual case where growth faltering comes from a recent illness or poor complementary feeding despite household advantage. The caregiver is told to continue the current diet instead of being referred, and the developmental window in which intervention is most effective for that child is missed.

3. Who bears the cost of errors?

The child and caregiver — not the clinic, the deploying organisation, or the data scientist who built the model. A missed stunted case costs a family a delayed intervention; a false positive costs, at most, an unnecessary follow-up assessment. Because these costs are so asymmetric, monitoring should prioritise the false-negative rate for "Severe stunting" and "Moderate stunting" specifically, not overall accuracy.

4. What redress mechanism exists?

None, currently. Kenya's Data Protection Act (KDPA) offers some recourse for data-protection violations but no general right of appeal against an automated decision in this setting. We propose that the model's output never function as a gate: any caregiver or health worker who suspects a "Normal growth" flag is wrong should be entitled to a manual anthropometric reassessment on request, with no referral or justification required.

5. How will the model be monitored?

We recommend quarterly re-evaluation against newly collected survey or facility growth-monitoring data, tracking macro-F1 and, specifically, per-class recall for "Severe stunting." If severe-stunting recall drops more than 10 percentage points below the value reported in the notebook's Part 7 evaluation, or any region's or residence-type's ROC-AUC falls more than 0.10 below the national figure, the model should be retrained or withdrawn from use in that subgroup until it is.

Fairness criterion

We apply Equal Opportunity: among children who are actually stunted, the model should correctly flag them at a similar rate regardless of region or residence type. Demographic Parity was rejected — stunting prevalence genuinely differs by region (see the notebook's §2.8), so forcing equal flag rates across regions would misstate real differences in risk. Equal Opportunity instead asks whether a stunted child's chance of being caught is similar no matter where they live.

Disaggregated findings

The notebook's §7.4 reports macro-F1 and ROC-AUC by residence type (urban / rural) and by region, for every subgroup with at least 20 test-set children. Overall test performance: macro F1 = 0.309, ROC-AUC (OvR) = 0.709. Largest residence-type ROC-AUC gap: 0.016. Largest region ROC-AUC gap: 0.475. This exceeds the 0.10 flag threshold and should be treated as a known, named limitation rather than folded into the overall figure.

Datasheet summary

The dataset is a nationally representative sample of Kenyan children under five from the most recent KDHS Children's Recode, but it undersamples remote pastoralist and insecure areas relative to their population share, and excludes households not reached during the survey window; populations outside the KDHS sampling frame — including refugee-camp residents and recent urban informal-settlement growth — are not represented.